

Data Foundations

Instructor-Led Course Syllabus

Format: Instructor-led; in person or virtual

Duration: 1 day (~7.5 hrs incl. breaks; ~6.25 hrs instruction)

Audience: New data learners across all roles

Prerequisite: Basic computer and spreadsheet familiarity

Course Description

Data Foundations is the on-ramp to every course in the Data Mastery catalog. In a single hands-on day, participants build the shared vocabulary and mental model that all data roles depend on: the business case for data, and the technical basics of how data is structured, stored, processed, and put to work. Short lectures alternate with guided activities and close with a practical capstone.

Learning Objectives

By the end of this course, participants will be able to:

- Explain the business case for data and why organizations invest in analytics
- Describe the data → information → insights chain
- Identify common data structures and data types
- Describe where data lives, including files (CSV, JSON, XML, binary) and databases
- Outline the data lifecycle from capture through analysis
- Map common data roles and tools to the right tasks and their learning path

Daily Agenda

Time	Module	Topics & Hands-On Activity
9:00–9:15	Welcome & Orientation	Course goals; the Data Mastery learning path; how this course connects to the tool-focused courses ahead.
9:15–10:15	1. The Business Case for Data	Why organizations invest in data; data → information → insights; drivers (revenue, cost, risk, customer); data in every role. Activity: identify a data use case from your own work.
10:15–10:30	Break	
10:30–11:45	2. What Data Is: Structure & Types	Basic data stream; structured / semi-structured / unstructured; data types (text, numeric, date, boolean); tables, rows, columns, granularity. Lab: read a real table and classify its columns.
11:45–12:30	Lunch	
12:30–1:30	3. Where Data Lives: Files & Databases	File types (CSV, JSON, XML, binary); relational databases; on-premise vs. cloud. Lab: open the same data as CSV and as a binary file and compare.
1:30–2:45	4. The Data Lifecycle	Capture, storage & transmission; preparation (nulls, naming); modeling (joins, unions); governance, security & quality; analytics & reporting. Activity: route a messy dataset through the lifecycle.
2:45–3:00	Break	
3:00–3:45	5. Roles, Tools & Your Learning Path	Data roles mapped to catalog volumes; tools (Excel, SQL, Python, R, Power BI, Tableau, Power Automate). Discussion: build your personal learning path.
3:45–4:30	Capstone & Wrap-Up	Guided capstone: explore and describe a real CSV end-to-end; knowledge-check review; Q&A; next steps and certificate.

Assessment & Materials

Assessment: Module knowledge checks plus a capstone exercise; an 80% threshold earns a certificate of completion.

Materials: Provided datasets, slide handouts, and free-trial tool links. Each participant needs a computer with internet and a spreadsheet application.